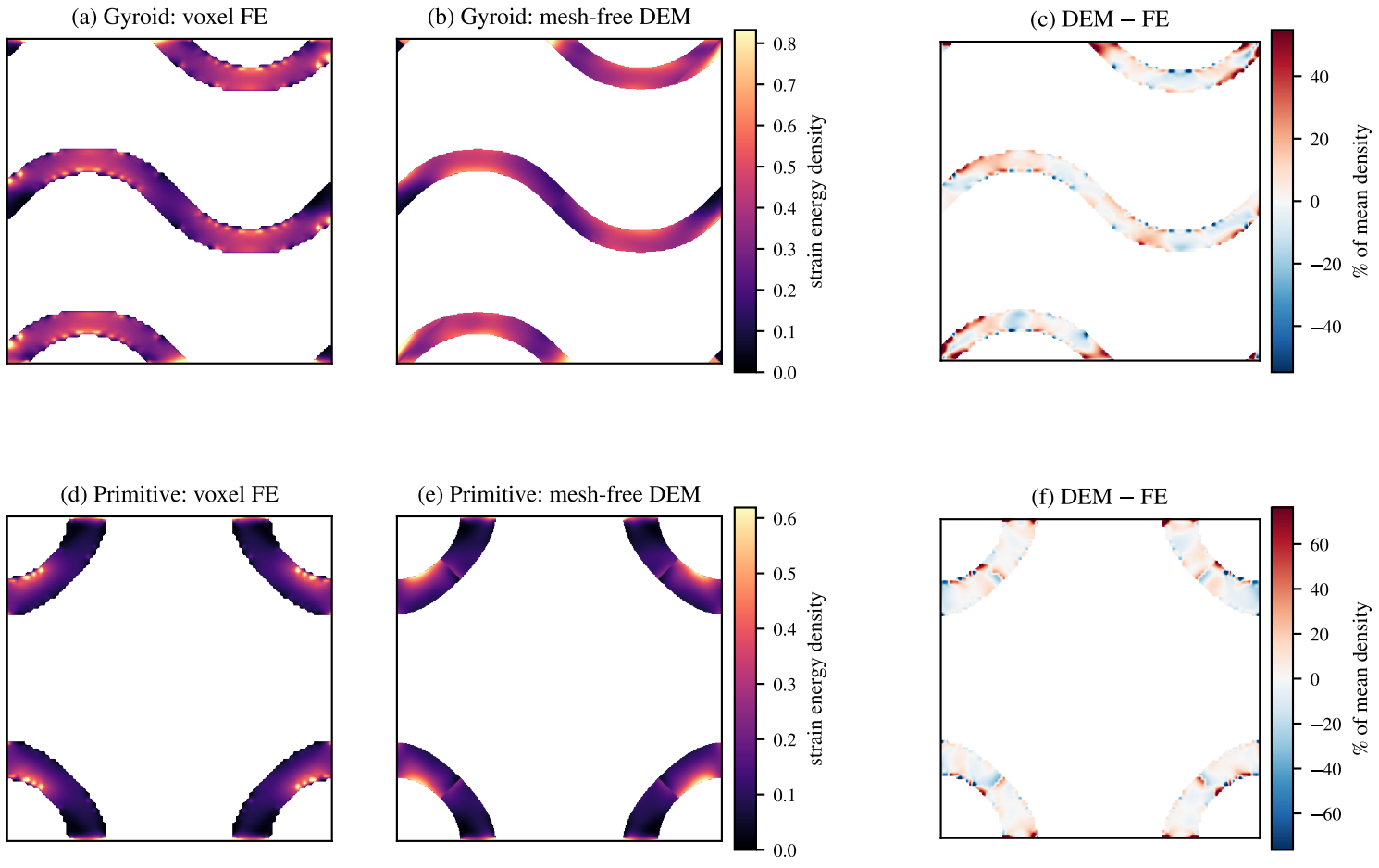

(a) Gyroid: voxel FE
(b) Gyroid: mesh-free DEM
(c) DEM − FE
strain energy density
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
% of mean density
−40
−20
0
20
40
(d) Primitive: voxel FE
(e) Primitive: mesh-free DEM
(f) DEM − FE
strain energy density
0.0
0.1
0.2
0.3
0.4
0.5
0.6
% of mean density
−60
−40
−20
0
20
40
60